

Bulan 8: Real-world Projects & Portfolio

Minggu 1: Project 1 - Healthcare Management System

Hari 1-3: Project Planning & Setup

Topic: Requirements analysis, architecture design, database schema

Proyek: MediCare - Healthcare Management API

Platform manajemen rumah sakit & klinik

Requirements Analysis:

- User stories & use cases
- Feature prioritization
- Architecture diagram
- Database schema design (ERD)
- API endpoint planning
- Technology stack selection

Setup:

- Django project initialization
- Docker setup
- PostgreSQL configuration
- Redis setup
- Celery configuration
- Environment configuration
- Git repository

Hari 4-5: Core Models & Authentication

Implementasi models & authentication

Models:

- CustomUser (Doctor, Nurse, Admin, Patient)
- Hospital (name, address, contact)
- Department (name, hospital, description)
- Doctor (user, specialization, department, schedule)

- Patient (user, medical_record_number, blood_type, allergies)
- Appointment (patient, doctor, datetime, status, type)
- MedicalRecord (patient, doctor, diagnosis, prescription, date)
- Prescription (record, medications, dosage, duration)

Features:

- JWT authentication
- Role-based access control
- User registration (patient, doctor)
- Email verification
- Password reset
- User profile management

Hari 6-7: API Development & Business Logic

REST API endpoints & business logic

Endpoints:

- /api/auth/* (register, login, verify, reset)
- /api/hospitals/* (CRUD for admins)
- /api/departments/* (CRUD, filter by hospital)
- /api/doctors/* (list, detail, by specialization, availability)
- /api/appointments/* (create, update, cancel, reschedule)
- /api/medical-records/* (CRUD, patient history)
- /api/prescriptions/* (create, list by patient)
- /api/patients/me/ (patient dashboard)
- /api/doctors/schedule/ (manage availability)

Business Logic:

- Appointment slot validation (no double booking)
- Doctor availability check
- Automated appointment reminders (Celery)
- Medical record access control (only assigned doctor & patient)
- Prescription validation

- Email notifications

Teknologi: DRF, JWT, Celery, Permissions

Minggu 2: Project 2 - Food Delivery Platform

Hari 8-10: Database Design & Core Features

Proyek: FoodExpress - Food delivery aggregator platform

Models:

- Restaurant (owner, name, address, cuisine_type, rating)
- MenuItem (restaurant, name, description, price, category, available)
- Cart (user, created_at)
- CartItem (cart, menu_item, quantity, customizations)
- Order (user, restaurant, items, total, status, delivery_address)
- OrderItem (order, menu_item, quantity, price)
- DeliveryDriver (user, vehicle_type, current_location, available)
- Delivery (order, driver, pickup_time, delivery_time, status)
- Review (user, restaurant, order, rating, comment)

Core Features:

- Restaurant registration & management
- Menu management (CRUD, categories)
- Real-time restaurant search (Elasticsearch)
- Cart functionality
- Order placement & tracking
- Payment integration (simulate Stripe)
- Driver assignment algorithm
- Real-time order tracking (WebSocket)

Hari 11-12: Advanced Features & Optimization

Advanced functionalities

Features:

- Geolocation-based restaurant search

- Filter (cuisine, rating, delivery time, price range)
- Popular dishes recommendation
- Order history
- Favorite restaurants
- Scheduled orders
- Promo codes & discounts
- Driver earnings tracking
- Restaurant analytics dashboard
- Multi-restaurant orders (separate deliveries)

Optimization:

- Cache popular restaurants
- Cache menu items
- Optimize geolocation queries (PostGIS)
- Database indexes for search
- Background job untuk analytics
- Image optimization untuk food photos

Teknologi: DRF, Elasticsearch, WebSocket, Celery, Redis, PostGIS

Hari 13-14: Testing & Deployment

Comprehensive testing & production deployment

Testing:

- Unit tests (models, serializers, business logic)
- API integration tests
- Authentication tests
- Permission tests
- Order flow tests
- Payment simulation tests
- 100+ test cases
- Coverage > 85%

Deployment:

- Docker multi-container setup
- PostgreSQL + PostGIS
- Redis cluster
- Celery workers
- Nginx load balancer
- SSL certificate
- CI/CD pipeline (GitHub Actions)
- Automated deployment
- Monitoring (Sentry, logging)

Documentation:

- API documentation (Swagger)
- Deployment guide
- Architecture documentation
- User guide

Teknologi: pytest, Docker Compose, CI/CD, Monitoring

Minggu 3: Project 3 - Learning Management System

Hari 15-17: Platform Setup & Content Management

Proyek: EduHub - Online learning platform

Models:

- Instructor (user, bio, expertise, rating)
- Course (instructor, title, description, category, price, level)
- Section (course, title, order)
- Lesson (section, title, content_type, video_url, duration)
- Enrollment (student, course, progress, completed)
- Quiz (lesson, title, passing_score)
- Question (quiz, question_text, question_type, points)
- Answer (question, answer_text, is_correct)
- StudentAnswer (student, question, answer, is_correct)
- Certificate (student, course, issued_date, certificate_url)

- Review (student, course, rating, comment)

Core Features:

- Instructor dashboard (create, manage courses)
- Course creation wizard (multi-step)
- Video upload & streaming (AWS S3 + CloudFront simulate)
- Course curriculum management (sections, lessons)
- Quiz creation & management
- Student enrollment
- Course player (video, documents, quizzes)
- Progress tracking
- Certificate generation (PDF)

Hari 18-19: Learning Experience & Gamification

Advanced learning features

Features:

- Adaptive learning path
- Discussion forum per course
- Live class scheduling (Zoom integration simulate)
- Assignment submission & grading
- Peer review system
- Gamification (badges, points, leaderboards)
- Learning analytics
- Personalized recommendations
- Notes & bookmarks
- Downloadable resources
- Offline mode (mobile app consideration)
- Subtitle support untuk videos

Social Features:

- Student profiles
- Instructor profiles
- Follow instructors

- Course reviews & ratings
- Q&A section
- Study groups

Admin Features:

- Course approval workflow
- Revenue sharing calculation
- Platform analytics
- User management
- Content moderation

Hari 20-21: Payment & Analytics

Payment processing & comprehensive analytics

Payment Features:

- Course purchase (Stripe simulate)
- Subscription plans (monthly, annual)
- Coupon codes
- Instructor payouts
- Refund handling
- Invoice generation

Analytics:

- Student progress analytics
- Course completion rates
- Quiz performance analytics
- Video engagement metrics
- Revenue reports
- Instructor performance
- Popular courses
- User retention metrics
- A/B testing framework

Performance:

- Video CDN optimization
- Image optimization
- Lazy loading
- Caching strategy (popular courses)
- Database query optimization
- API response time optimization

Teknologi: DRF, Video streaming, PDF generation, Stripe API, Analytics, Caching

Minggu 4: Portfolio & Job Preparation

Hari 22-23: Portfolio Website Development

Personal portfolio & project showcase

Portfolio Website Features:

- About me section
- Skills showcase (Django, DRF, Docker, etc.)
- Project gallery (3 major projects)
- Blog untuk technical articles
- Contact form
- Resume download
- Responsive design
- SEO optimization

Project Documentation:

- Detailed README untuk setiap project
- Architecture diagrams
- API documentation
- Setup instructions
- Screenshots & demos
- Video walkthroughs
- Live demo links
- GitHub repositories (clean code, good commits)

Hari 24: Resume & LinkedIn Optimization

Professional branding

Resume:

- Backend Developer resume
- Highlight Django projects
- Quantify achievements (reduced API latency by 40%, etc.)
- Technology stack showcase
- Certifications (if any)
- ATS-friendly format

LinkedIn:

- Optimize profile (headline, about, experience)
- Add projects dengan descriptions
- Skills endorsements
- Recommendations
- Post technical content
- Connect with Django community

GitHub:

- Clean GitHub profile
- Pinned repositories (best projects)
- Contribution graph
- README profile
- Consistent commit history

Hari 25-26: Interview Preparation

Technical interview practice

Topics to Review:

- Django fundamentals Q&A
- DRF concepts
- Database optimization

- System design questions
- Authentication & security
- API design best practices
- Deployment & DevOps
- Problem-solving (algorithms)

Practice:

- Mock technical interviews
- System design scenarios
- Coding challenges
- Behavioral questions
- Project walkthrough practice

Preparation:

- Common Django interview questions (50+)
- Design patterns explanations
- Scalability discussions
- Trade-offs in architecture decisions

Hari 27-28: Job Applications & Final Polish

Launch job search

Job Search Strategy:

- Identify target companies
- Job board exploration (LinkedIn, Indeed, Glassdoor)
- Company research
- Customize resume per application
- Write compelling cover letters

Final Polish:

- Code review all projects
- Fix any bugs
- Update documentation
- Improve test coverage

- Performance optimization
- Security audit
- Deploy all projects
- Create project demo videos

Applications:

- Apply to 10+ positions
- Follow up emails
- Network with recruiters
- Attend Django meetups (virtual)
- Contribute to open source (Django projects)

Continuous Learning:

- Advanced Django topics wishlist
- Next technologies to learn
- Contributing to Django community
- Building personal brand

8 Month Journey Complete! 🎉

Total Achievements: ✅ 200+ hands-on projects completed ✅ 12+ major production-ready applications ✅ 3 portfolio-worthy capstone projects:

- Healthcare Management System
- Food Delivery Platform
- Learning Management System ✅ Full-stack backend expertise ✅ Production deployment experience
✅ CI/CD pipeline mastery ✅ Testing & quality assurance ✅ System design & scalability knowledge
✅ Professional portfolio ✅ Job-ready resume ✅ Interview preparation complete

Technology Mastery:

- Python & Advanced Python
- Django & Django REST Framework
- PostgreSQL & Database optimization
- Redis & Caching

- Celery & Background tasks
- Docker & Containerization
- CI/CD & GitHub Actions
- Nginx & Deployment
- Testing & Quality assurance
- System design & Architecture patterns
- Security best practices
- Performance optimization

You are now a job-ready Django Backend Developer! 🚀

Next Steps:

1. Keep building & learning
2. Contribute to open source
3. Network with developers
4. Apply for jobs confidently
5. Continue growing your skills

Remember: The journey doesn't end here. Keep coding, keep learning, keep growing! Good luck with your Django career! 💪